

## Python List Methods Examples

Python List `append()` Method with ExamplesPython List `clear()` Method with ExamplesPython List `copy()` Method with ExamplesPython List `insert()` Method with ExamplesPython List `pop()` Method with ExamplesPython List `remove()` Method with ExamplesPython List `reverse()` Method with ExamplesPython List `sort()` Method with ExamplesPython List `count()` method with ExamplesPython List `extend()` Method with ExamplesPython List `index()` Method with Examples

## 1.Count the Number of Null elements in a List

```
gvn_lst = ["btechgeeks", "hello", "", 123, ""]
cou_nt = 0

for itetor in range(len(gvn_lst)):
    if(gvn_lst[itetor] == ""):
        cou_nt += 1

print("The Number of Null Elements in the above given list = ", cou_nt)
```

## 2.Print Items from a List with Specific Length

```
gvn_lst = ['hello', 'btechgeeks', 'good', 'morning']
gvn_len = 5

print("The Items from a given List with given Specific Length : ")
for itr in gvn_lst:
    if(len(itr)) == gvn_len:
        print(itr)
```

Output: The Items from a given List with given Specific Length : hello

## 3.Find the Second Largest Number in a List

```
given_list = [8, 12, 4, 6, 7, 3, 2, 6, 1, 9]

given_list.sort(reverse=True)
seclarg = given_list[1]

print("printing the second largest element in the given list : ", seclarg)
```

#### 4.Swap the First and Last Value of the Given List

```
given_list = ['hello', 'this', 'is', 'BTechGeeks', 'online', 'platform', 'for', 'btech', 'students']
print('printing the given list before swapping the values = ',given_list)

# Swap the first and last elements in the given list using a , operator
given_list[0], given_list[-1] = given_list[-1], given_list[0]

print('printing the given list after swapping the values = ',given_list)

//or

given_list = [33, 92, 12, 74, 22, 79, 122, 17, 63, 99, 129]
print('printing the given list before swapping the values = ',given_list)

tempo = given_list[0]
given_list[0] = given_list[-1]
given_list[-1] = tempo

print('printing the given list after swapping the values = ',given_list)
```

#### 5.Calculate the Average of Numbers in a Given List

```
given_list = [8, 9, 1, 23, 15, 20, 19, 13, 8, 7, 5, 2, 7, 10, 14, 16]
sumOfList = 0

* for eleme in given_list:
    sumOfList = sumOfList+eleme

length = len(given_list)
```

```
listAvg = sumOfList/length
print("The average value of the given list", given_list, "=", listAvg)
```

```
*
given_list = [8, 9, 1, 23, 15, 20, 19, 13, 8, 7, 5, 2, 7, 10, 14, 16]
sumOfList = sum(given_list)
length = len(given_list)
listAvg = sumOfList/length
print("The average value of the given list", given_list, "=", listAvg)
```

6. Print Sum of Negative Numbers, Positive Even Numbers and Positive Odd numbers in a List

```
given_list = [23, 128, -4, -19, 233, 726, 198, 199, 203, -13]
```

```
posEven = []
posOdd = []
negNum = []
```

```
for element in given_list:
```

```
    if(element > 0):    # checking if the number is greater than 0
```

```
        if(element % 2 == 0):
            posEven.append(element)
        else:
            posOdd.append(element)
```

```
    else:
        negNum.append(element)    # else if the number is less than 0 then add to negNum list using append() function
```

```
# Calculating sum
posEvenSum = sum(posEven)
posOddSum = sum(posOdd)
negNumSum = sum(negNum)
```

```
print("The sum of all positive even numbers in the given list ", given_list, "=", posEvenSum)
```

```
print("The sum of all positive odd numbers in thee given list ", given_list, "=", posOddsum)
print("The sum of all positive negative numbers in thee given list ", given_list, "=", negNumsum)
```

output:

```
The sum of all positive even numbers in thee given list [23, 128, -4, -19, 233, 726, 198, 199, 203, -13] = 1052
The sum of all positive odd numbers in thee given list [23, 128, -4, -19, 233, 726, 198, 199, 203, -13] = 658
The sum of all positive negative numbers in thee given list [23, 128, -4, -19, 233, 726, 198, 199, 203, -13] = -36
```

### 7.Print Largest Even and Largest Odd Number in a List

```
* given_list = [532, 234, 9273, 845, 1023, 9]
```

```
evenNumList = []
oddNumList = []
```

```
for eleme in given_list:
```

```
    if(eleme % 2 == 0):
        evenNumList.append(eleme)
    else:
        oddNumList.append(eleme)
```

```
evenNumList.sort()
oddNumList.sort()
```

```
print("The Largest even number in the given list", given_list, "=", evenNumList[-1])
print("The Largest odd number in the given list", given_list, "=", oddNumList[-1])
```

Output:

```
The Largest even number in the given list [532, 234, 9273, 845, 1023, 9] = 532
The Largest odd number in the given list [532, 234, 9273, 845, 1023, 9] = 9273
```

```
* given_list = [532, 234, 9273, 845, 1023, 9]
```

```
evenNumList = [eleme for eleme in given_list if eleme % 2 == 0]
oddNumList = [eleme for eleme in given_list if eleme % 2 != 0]      #List Comprehension
```

```
evenNumList.sort()
oddNumList.sort()
print("The Largest even number in the given list", given_list, "=", evenNumList[-1])
print("The Largest odd number in the given list", given_list, "=", oddNumList[-1])
```

```
* #List Comprehension(User Input)
totalCount = int(input("Enter the total number of elements of the given list = "))

given_list = []

for i in range(totalCount):
    eleme = int(input("Enter some random element(integer) = "))
    given_list.append(eleme)

evenNumList = [eleme for eleme in given_list if eleme % 2 == 0]
oddNumList = [eleme for eleme in given_list if eleme % 2 != 0]

evenNumList.sort()
oddNumList.sort()
print("The Largest even number in the given list", given_list, "=", evenNumList[-1])
print("The Largest odd number in the given list", given_list, "=", oddNumList[-1])
```

#### 8.Add Number to each Element in a List

```
gven_lst = [2,4,6,8,10]
numbr = 5
len_lst = len(gven_lst)

for i in range(len_lst):
    gven_lst[i] = gven_lst[i]+numbr

print("The given list after addition of a number to each element in a list = ", gven_lst)
```

Output:

The given list after addition of a number to each element in a list = [7, 9, 11, 13, 15]

```
# Give the List as user input using list(),map(),input(),and split() functions
gven_lst = list(map(int, input('Enter some random List Elements separated by spaces = ').split()))
numbr = int(input('Enter some random number = '))
len_lst = len(gven_lst)

for i in range(len_lst):
    gven_lst[i] = gven_lst[i]+numbr

print("The given list after addition of a number to each element in a list = ", gven_lst)
```

#### 9.Divide all Elements of a List by a Number

```
gven_Lst = [3, 2.5, 42, 24, 15, 32]
gvn_numbr = 2
Lst_aftr_div = []

for itr in gven_Lst:
    Lst_aftr_div.append(itr/gvn_numbr)

print("The above given list after division all elements of a List by the given Number=", Lst_aftr_div)
```

Output:

The above given list after division all elements of a List by the given Number= [1.5, 1.25, 21.0, 12.0, 7.5, 16.0]

#### 10.Move all Zeros present in an Array/List to the End

```
given_list = [7, 2, 0, 34, 56, 12, 0, 5, 6, 8, 0, 0, 9, 0, 1, 2, 4, 5]
print("the list before modification :")
print(given_list)

index = 0
for eleme in range(len(given_list)):
    if(given_list[eleme] != 0):
        given_list[index], given_list[eleme] = given_list[eleme], given_list[index] # swappng index element with current index
        index = index+1
```



```
print("the list after modification :")
print(given_list)
```

Output:

```
the list before modification : [7, 2, 0, 34, 56, 12, 0, 5, 6, 8, 0, 0, 9, 0, 1, 2, 4, 5]
the list after modification : [7, 2, 34, 56, 12, 5, 6, 8, 9, 1, 2, 4, 5, 0, 0, 0, 0, 0]
```

11. Find the Missing Number in an array/list

```
def findMissingNumb(given_list):

    n = len(given_list)

    length = n + 1      # Because a number is missing from the list, the real size is n+1.

    # Getting sum of all numbers from 1 to length using mathematical formula
    totalSum = length * (length + 1) / 2
    arraysum = 0

    for ele in given_list:
        arraysum = arraysum+ele

    missnum = totalSum-arraysum

    return int(missnum)

given_list = [1, 3, 2, 6, 5]
print("The missing number in the given list", findMissingNumb(given_list))
```

12. Put Even and Odd elements in a List into Two Different Lists

```
given_list = [7, 24, 72, 39, 65, 87, 93, 27, 64, 96, 82, 36, 47, 75, 12, 58, 97]
evenList = []
oddList = []

for element in given_list:
    if(element % 2 == 0):
        evenList.append(element)
```

```
    else:
        oddList.append(element)

print("The even elements present in given list are :")
print(evenList)
print("The odd elements present in given list are :")
print(oddList)
```

Output:

The even elements present in given list are : [24, 72, 64, 96, 82, 36, 12, 58]  
The odd elements present in given list are : [7, 39, 65, 87, 93, 27, 47, 75, 97]

### 13. Find the Largest Number in a List

```
gvnList = [12, 34, 11, 9, 2, 4, 1, 23, 28, 29, 25, 23, 31, 18, 15, 8, 13, 15, 17]

gvnList.sort() #sorts the elements in ascending order
lstLeng = len(gvnList)
largstele = gvnList[lstLeng-1]
print('The largest element present in the given list', gvnList, '=', largstele)
```

### 14. Get the Position of Max Value in a List

```
Gvn_lst = [1, 5, 9, 2, 7, 3, 8]
maxim_vle = max(Gvn_lst)
print("Maximum Value in the above Given list = ", maxim_vle)

maxim_positn = Gvn_lst.index(maxim_vle)
print("Position of Maximum value of the above Given List = ", maxim_positn+1) # ( since list index starts from zero).
```

### 15. Print Number in Ascending Order which contains 1, 2 and 3 in their Digits

```
gven_lst = [67123, 1234, 985, 126, 1011]

numb = []
```



```
for itr in gven_lst:
    strng_number = str(itr)
    if '1' in strng_number and '2' in strng_number and '3' in strng_number:
        numb.append(itr)
```

```
numb.sort()
print("The numbers with the digits 1, 2, and 3 in ascending order, which is separated by commas= ", numb)
```

Output: The numbers with the digits 1, 2, and 3 in ascending order, which is separated by commas= [1234, 67123]

#### 16. Get Sum of all the Factors of a Number

```
gvn_numb = 24

all_factors = [1]
for itr in range(2, gvn_numb+1):
    if gvn_numb % itr == 0:

reslt = sum(all_factors)
print("The Sum of all the factors of {", gvn_numb, "} is = ", reslt)
```

Output: The Sum of all the factors of { 24 } is = 60

#### 17. Find the Sum of Even Factors of a Number

```
gvn_numb = 24

all_factors = []

for itr in range(1, gvn_numb+1):
    if gvn_numb % itr == 0:
        if itr % 2 == 0:

reslt = sum(all_factors)
print("The Sum of all even factors of {", gvn_numb, "} = ", reslt)
```

Output: The Sum of all even factors of { 24 } = 56

tuples:==

Create a List of Tuples with the First Element as the Number and Second Element as the Square of the Number

```
lowerlimit = 5
upperlimit = 13

listOfTuples = [(k, k**2) for k in range(lowerlimit, upperlimit+1)]
print('The list of Tuples are : ', listOfTuples)
```

Output: The list of Tuples are :  
[(5, 25), (6, 36), (7, 49), (8, 64), (9, 81), (10, 100), (11, 121), (12, 144), (13, 169)]

Remove Elements from a Tuple

```
gven_tup = (14, 32, 16, 85, 47, 65)
num = 3

fst_part = gven_tup[:num] # Do out the Slicing from '0 ' to given 'n-1' and store it in a variable.
secnd_part = gven_tup[num+1:]
gven_tup = fst_part + secnd_part

print("The above given tuple after removal of {", num, "} Element = ", gven_tup)
```

Output: The above given tuple after removal of { 3 } Element = (14, 32, 16, 47, 65)

Python size of tuple

```
gvntuple = (9, 11, 24, 19, 11, 23, 29, 23, 31)

tuplelength = len(gvntuple)
//or
tuplelength = 0
for elemen in gvntuple:
    tuplelength = tuplelength+1
```

```
print('The length of the given tuple', gvn_tuple, 'is [', tuplelength, ']')
```

Output: The length of the given tuple (9, 11, 24, 19, 11, 23, 29, 23, 31) is [ 9 ]

Find Sum of Tuple Items

```
gvn_tuple = tuple(map(int, input( 'Enter some random tuple Elements separated by spaces = ').split()))
```

```
sum_tuple = 0
```

```
for elemnt in gvn_tuple:  
    sum_tuple = sum_tuple + elemnt
```

```
print("The Sum of the given tuple elements", gvn_tuple, "=", sum_tuple)
```

get the First Element of each Tuple in a List

```
gvn_tuple_list = [(1, 2), (3, 4)]
```

```
fst_tuple_elems = []
```

```
for elemnt in gvn_tuple_list:  
    fst_tuple_elems.append(elemnt[0])
```

```
print("The first element of the each tuple in a given tuplelist", gvn_tuple_list, "=", fst_tuple_elems)
```

Output: The first element of the each tuple in a given tuplelist [(1, 2), (3, 4)] = [1, 3]